

# HYEONBEEN LEE

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## EDUCATION

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**Virginia Tech**, Blacksburg, VA, United States

Aug. 2026 — May 2031 (Expected)

Ph.D. in Mechanical Engineering (Advisor: [Simon Stepputtis](#))Research Focus: Vision-Language-Action (VLA), Contact-rich manipulation, Real-time controller**Kyung Hee University**, Yongin, South Korea

Mar. 2022 — Feb. 2024

M.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#))

GPA: 3.93/4.0

Thesis: ‘Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems’**Kyung Hee University**, Yongin, South Korea

Mar. 2015 — Feb. 2022

B.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#), [Shinkyu Jeong](#))

(Upper-Division: 3.88/4.0) GPA: 3.52/4.0

Thesis: ‘Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization’

(\* Military leave of absence: May 2017 — Feb. 2019)

## PUBLICATIONS

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### Journal Papers

4. **H. Lee**, M. Jung, T.K. Yeu, J.B. Han, D. Park, S. Stepputtis, J.G. Kim (2026), ‘Frequency-aware decomposition learning for sensorless wrench estimation in vibration-rich robotic contact’, submitted to *Robotics and Computer-Integrated Manufacturing*. doi:10.48550/arXiv.2604.12905 [\[preprint\]](#) [\[code\]](#) [\[dataset\]](#)
3. B. Koo, M. Jung, **H. Lee**, T. Yeu, J.G. Kim, J.B. Han, Y. Lee, D. Park (2026), ‘Buoyancy-integrated hybrid reaction force estimation method with real-time haptic feedback for underwater hydraulic manipulation’, revised from *IEEE Robotics and Automation Letters*.
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), ‘cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics’, *Journal of Computational Physics (WoS IF Top 2.5% in Physics, Mathematical)*, 496, 112578. doi: 10.1016/j.jcp.2023.112578 [\[paper\]](#) [\[code\]](#)
1. S. Han\*, G.E. Jeong\*, **H. Lee**, W.S. Choi, J.G. Kim (2023), ‘Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions’, *Nuclear Engineering and Technology (WoS IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. (\*Co-first authors) doi: 10.1016/j.net.2023.07.033 [\[paper\]](#)

### Conference Proceedings & Presentations

6. J.H. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu, ‘Digital twin model of underwater construction robot for real-time grinding simulation’, *7th International Conference on Multibody System Dynamics (IMSD)*, Madison, WI, United States, Jun. 2024. [\[abstract\]](#)
5. **H. Lee**, J. Han, T.K. Yeu, J.G. Kim, ‘Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer’, *Korean Society of Mechanical Engineers (KSME)* (Oral Presentation), Incheon, South Korea, Nov. 2023. [\[abstract\]](#)
4. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation’, *Korean Society of Mechanical Engineers (KSME)* (**Excellent Paper Award**, Oral Presentation, \* Extended works from IMAC 2023), Busan, South Korea, May 2023. [\[abstract\]](#)
3. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Composite neural network with differentiation propagation for modeling nonlinear impulsive dynamics’, *Conference on Dynamics and Control in Korean Society of Mechanical Engineers (KSME)* (Oral Presentation), Jeju, South Korea, Mar 2023. [\[abstract\]](#)
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Composite neural network framework for modeling impulsive nonlinear dynamic responses’, *41st International Modal Analysis Conference (IMAC-XLI)* (Oral Presentation), Austin, TX, United States, Feb. 2023. doi: 10.1007/978-3-031-34946-1.21 [\[paper\]](#)
1. **H. Lee**, J.G. Kim. ‘Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation’, *Korean Society for Noise and Vibration Engineering (KSNVE)* (Oral Presentation), Jeju, South Korea, Dec 2022.

## AWARDS & HONORS

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- **Excellent Paper Award** Korean Society of Mechanical Engineers, 2023
  - Awarded for a first-author presentation at KSME 2023 conference.
- **Brain Korea 21 Plus Scholarship** Graduate School, Kyung Hee University, 2022—2023
  - Received \$18,000 during M.Eng. studies through a national graduate research scholarship.
- **Graduate Assistantship** Dept. of Mechanical Engineering, Kyung Hee University, 2022—2023
  - Received \$12,000 during M.Eng. studies through a selective departmental assistantship.
- **Academic Superiority Scholarship** Dept. of Mechanical Engineering, Kyung Hee University, 2021
  - Ranked 1st in the departmental merit review and received a full tuition waiver.

## RESEARCH EXPERIENCES

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**Virginia Tech, Department of Mechanical Engineering** Graduate Research Assistant  
Thinking Embodied Agents Lab., advised by *Dr. Simon Stepputtis* Aug. 2026—Present

- Developing hierarchical Vision-Language-Action (VLA) systems and sensorless contact perception methods for real-time robot controllers.

**Kyung Hee University, Department of Mechanical Engineering** Research Assistant  
Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim* Mar. 2024—Apr. 2026

- Proposed a frequency-aware decomposition learning framework for sensorless wrench estimation in contact- and vibration-rich robotic contact. Submitted a first-authored manuscript.
- Optimized the developed wrench estimation model architectures for real-time closed-loop deployment with Korea Research Institute of Ships and Ocean Engineering (KRISO).

**KAIST, Kim Jaechul Graduate School of Artificial Intelligence** Research Intern  
Cognitive Learning for Vision & Robotics Lab., advised by *Dr. Joseph J. Lim* Sep. 2023—Nov. 2023

- Investigated skill-prior regularization method for accelerating reinforcement learning in long-horizon robotic manipulation.

**Kyung Hee University, Department of Mechanical Engineering** Graduate Research Assistant  
Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim* Mar. 2022—Feb. 2024

- Proposed a composite neural network architecture for learning impulsive high-order dynamics from multi-order derivatives. Published a first-author article in the *Journal of Computational Physics*.
- Developed a multibody dynamics vehicle model for the Korea Atomic Energy Research Institute (KAERI). Published a co-authored article in the *Nuclear Engineering and Technology*.
- Designed a data-driven model for sensorless wrench estimation in a 6-DoF hydraulic robot grinding for the Korea Research Institute of Ships and Ocean Engineering (KRISO).

**Kyung Hee University, Department of Mechanical Engineering** Undergraduate Research Assistant  
Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim* Jan. 2021—Feb. 2022  
Aerodynamic Design Exploration & Big Data Analysis Lab, advised by *Dr. Shinkyu Jeong* Sep. 2020—Jun. 2021

- Investigated data-driven methods for flexible multibody dynamics and aerodynamic design optimization and led an undergraduate thesis project.

## INVITED SEMINARS

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2. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, IMBD Lab, Chungnam National University, Daejeon, South Korea, Apr. 16, 2026 [\[keynote\]](#)
1. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, Modeling & Simulation Lab, Kyung Hee University, Yongin, South Korea, Apr. 14, 2026 [\[keynote\]](#)

## TEACHING & MENTORSHIP

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**Research Mentor** Modeling & Simulation Lab., Kyung Hee University  
— Guided junior graduate students on data-driven modeling and robotics. Mar. 2025—Jul. 2026

**Teaching Assistant, ME380 Systems Dynamics** Dept. of Mechanical Engineering, Kyung Hee University  
— Lectured on multibody dynamics with theoretical derivation and its computational implementation. Mar. 2022—Jun. 2023

## COMMUNITY SERVICE

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**Lead Administrative Assistant**  
**48th College of Engineering Student Council**  
**ROK-U.S. Combined Marine Corps Interpreter**

Dept. of Mechanical Engineering, Kyung Hee University, 2022—2024  
Kyung Hee University, Feb 2019—Jan. 2020  
1st Marine Div., ROKMC, Sep 2017—Feb. 2019

## REFERENCES

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- **Jin-Gyun Kim**, Ph.D.

Associate Professor at Department of Mechanical Engineering, Kyung Hee University ([jingyun.kim@khu.ac.kr](mailto:jingyun.kim@khu.ac.kr))

- **Hee-Sun Choi**, Ph.D.

Assistant Professor at Department of Artificial Intelligence and Robotics, Sejong University ([heesunchoi@sejong.ac.kr](mailto:heesunchoi@sejong.ac.kr))

- **Seongji Han**, Ph.D.

Assistant Professor at Department of Mechatronics Engineering, Chungnam National University ([seongji.han@cnu.ac.kr](mailto:seongji.han@cnu.ac.kr))